SAI NIHAL DIDDI

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EDUCATION

UNIVERSITY OF VICTORIA — **Victoria, BC**

BSc Computer Science - Specialization in Software Systems — *Sep 2021 - Dec 2026*

Current Senior-Level GPA: ***3.82/4.0***

Achievements: *UVic INSPIRE Hackathon Winner - 2025*

WORK EXPERIENCE

SOFTWARE DEVELOPMENT & AUTOMATION - CO-OP — **Vancouver, BC**

Motorola Solutions — *Jan 2026 - Aug 2026*

- Optimized C++ token-authentication services by restructuring database call logic, reducing LDAP queries from over 1,000 to one per request and lowering directory-service load.
- Developed Python tools to convert system crash dumps into readable diagnostic logs, helping developers investigate system failures.
- Built AI agents to help resolve UI element locator issues in the automated testing framework.
- Helped develop AI agents to migrate end-to-end tests from a legacy Ruby framework to a new Python and Selenium framework, covering UI features, hardware panels, and LDAP schema validation.
- Collaborated with the software development team to build and ship new UI components in a React.js and Ruby application.

ML RESEARCH ASSISTANT — **Victoria, BC**

University of Victoria — *May 2025 - Nov 2025*

- Investigated speculative decoding for faster LLM inference, evaluating token-verification batch sizes, dynamic block sizing, and latency trade-offs.

JUNIOR QA ANALYST - CO-OP — **Victoria, BC**

Journal Technologies — *Jan 2024 - Aug 2024*

- Built reusable page-object models using Java and Selenium WebDriver, increasing automation coverage by 30% and reducing repetitive test writing.
- Integrated automated regression checks into CI/CD pipelines and worked with developers to investigate flaky tests across deployments.
- Analyzed application logs and helped resolve bugs in eCourt and eProsecutor, working with developers and product owners to prioritize issues.
- Migrated test management from Excel to TestRail, reducing duplicated test cases and improving traceability between requirements, tests, and defects.

PROJECTS

REALTIME FRAUD DETECTION SYSTEM | Python, Pandas, NumPy, Kafka, Apache Spark, scikit-learn, Docker, Streamlit **|** GitHub Link

- Built a real-time fraud detection pipeline using Kafka and PySpark Structured Streaming, supporting Isolation Forest anomaly detection and supervised classification with XGBoost or Logistic Regression.
- Sent fraud alerts through Kafka, stored them in PostgreSQL, and displayed alert metrics in a Streamlit dashboard, using saved model settings to keep scoring consistent.
- Tested PostgreSQL storage with 3,000 writes for 1,000 simulated transactions, confirming no duplicate records and preserving saved alerts after a database restart in local tests.

LUCENT - AI-POWERED NOTE MANAGEMENT | React Native, Expo, Gemini API, Postgres, Docker, FastAPI **|** Web App **|** GitHub Link

- Built a note-management app with a shared web, iOS, and Android codebase that turns typed and handwritten notes into summaries, tasks, and topic tags.
- Added handwriting input, Gemini transcription, related-note retrieval, and PostgreSQL storage; deployed the web frontend on GitHub Pages and backend on Vercel.

MULTI-AGENT FEATURE FLAG MONITORING | Go, Python, FastAPI, PostgreSQL, Redis, Gemini API, Docker **|** GitHub Link

- Built a system that assigns devices to new or existing features and uses Python agents to compare failures, recommend disabling features or further review, and investigate feature conflicts using simulated data.
- Tested routing across 100,000 simulated devices in local tests, confirming consistent assignments on repeat checks and no devices switching back to the old version when rollout increased from 25% to 50%.

SKILLS

Programming Languages: Python, Java, C, SQL, JavaScript/TypeScript, Ruby, Go

Frameworks & Libraries: FastAPI, PySpark, Selenium, scikit-learn, React, Node.js

Cloud, Data & DevOps: Docker, AWS (EC2, RDS, S3), Kafka, PostgreSQL, LDAP, CI/CD (Jenkins, GitHub Actions, Bitbucket)

AI Assisted Development tools: OpenAI Codex, Claude Code, GitHub Copilot